CRC Liver-Metastasis Invasion-Front Model

Interview and Viva Questions with Model Answers

QUICK FACTS	22,260 Visium spots | 6 sections | 2 liver sections | ROC AUC 0.908 | permutation P 0.0196

How to use this guide

Answer in three layers: first state the design decision, then give the executed evidence, then name the limitation or next validation. The model answers below are intentionally concise enough for interviews but technically precise enough for a viva.

Concept and data

Question 1. What problem does the project solve?

It models the tumour-liver interface directly. Standard spot-level or whole-slide analysis can merge malignant epithelial plasticity with fibroblast, macrophage and hepatocyte RNA. The workflow preserves compartments and spatial direction so candidate invasion mechanisms can be compared.

Question 2. What is the central hypothesis?

Partial-EMT CRC cells occupy distinct hepatocyte-contact or fibroblast-rich ecosystems, with separable macrophage, endothelial, ECM, glycolytic and hypoxic gradients. Typed spatial context should transfer between liver sections better than unstructured spot features.

Question 3. Why focus on the invasion front?

Histopathological growth patterns are defined at the tumour-liver boundary. Desmoplastic growth has a fibrotic rim; replacement growth brings cancer cells into contact with liver plates and may use vessel co-option. Whole-slide averages can erase this biology.

Question 4. Which public data were used?

GSE225857 supplied four primary CRC and two liver-metastasis Visium FFPE sections, totaling 22,260 spots, plus processed single-cell tables. Zenodo 7520789 supplied an open 48-AOI GeoMx count table used as an orthogonal evidence asset.

Question 5. How was the single-cell reference built?

Author-annotated processed cells were harmonized and aggregated into pseudobulk malignant CRC, activated fibroblast, macrophage/myeloid and endothelial profiles. The recovered groups contained 56,072 cells. Hepatocytes were absent and were reported as unavailable rather than fabricated.

Methodology and technical design

Question 6. How were EMT and partial EMT scored?

Counts were normalized within section. Curated epithelial, mesenchymal and CRC-lineage programs were converted to within-section percentile scores. Partial EMT requires concurrent epithelial, mesenchymal and malignant-lineage support.

Question 7. How did you prevent stromal contamination from becoming malignant EMT?

Mesenchymal-high, malignant-low and fibroblast-high spots receive a stromal-inflation warning. Only CRC-lineage-supported spots can enter the malignant partial-EMT compartment or the front/core model.

Question 8. How was the tumour-liver interface defined?

A six-neighbour spatial graph was built from Visium array coordinates. High-confidence malignant-hepatocyte contacts seeded the boundary. Breadth-first graph distance then assigned tumour-front and liver-front spots for three hops, approximately 300 micrometres.

Question 9. Why approximately 300 micrometres?

Three Visium centre-to-centre graph steps are roughly 300 micrometres and mirror the tumour- and liver-front width used in recent pathologist-reviewed CRLM spatial work. It remains an approximation that should be sensitivity-tested.

Question 10. What makes the graph heterogeneous?

Nodes carry cell-compartment labels. Edges distinguish physical proximity, compartment contact, direct tumour-hepatocyte contact, ligand-receptor co-expression, and directional ECM or glycolysis gradients. Different relations are aggregated separately.

Question 11. Which ligand-receptor axes are represented?

The hypothesis set includes CCL18-PITPNM3, SPP1-CD44/ITGAV, TGF-beta-TGFBR2, CXCL12-CXCR4, HGF-MET, VEGFA-KDR, ANGPT2-TEK, COL1A1-ITGA2 and GAS6-AXL. These edges indicate co-expression opportunities, not functional signalling.

Question 12. Is the method a GNN?

V1 performs typed message passing but uses standardized logistic regression as the final classifier. This is a graph-informed baseline rather than a deep relational GNN. With only two independent liver sections, the simpler model is easier to audit and less prone to overfitting.

Question 13. How was train-test leakage controlled?

Whole liver sections, not random spots, were held out. Boundary-defining cross-interface messages were excluded from predictive features. Labels were permuted within section for the null comparison.

Results and interpretation

Question 14. What was the model performance?

Across leave-one-liver-section-out predictions, ROC AUC was 0.908, average precision 0.857 and balanced accuracy 0.831. The empirical P value from 50 within-section permutations was 0.0196; null mean AUC was 0.495.

Question 15. Why does AUC 0.908 not prove clinical value?

The endpoint is an inferred weak spatial label, only two sections are independent biological units, and thousands of spots are not thousands of patients. The metric demonstrates technical transfer between these sections, not prognosis, treatment response or population generalization.

Question 16. What was the most important positive result?

Damaged-hepatocyte context was higher at the inferred front, and neighbouring damaged-hepatocyte and MRC1/CCL18-like macrophage programs were among the strongest front-associated model features. This prioritizes a host-injury and myeloid-contact hypothesis.

Question 17. What was the most important negative result?

Partial EMT, glycolysis and hypoxia were not uniformly higher at the inferred tumour front. They were more prominent in malignant core regions. That rejects a simplistic invasive-ring story and changes the validation plan.

Question 18. How do you interpret the front-versus-core architecture?

The current data are consistent with a host-contact front surrounding a metabolically stressed, pEMT-rich core. Competing explanations include section-specific growth architecture, Visium spot mixing and imperfect weak labels. Pathology and protein validation are needed to distinguish them.

Question 19. What do desmoplastic and replacement growth patterns mean here?

Desmoplastic growth implies a fibroblast/ECM-rich rim separating tumour and liver. Replacement growth implies direct malignant-hepatocyte contact and possible vessel co-option. Both sections remained mixed or uncertain proxies because H&E; was not scored by pathologists.

Question 20. What are the project’s main methodological advantages?

It is compartment-aware, spatially directional, relation-specific, low-capacity, leakage-controlled and reproducible. It reports missing evidence instead of silently imputing it and exposes unsupported claims in the UI.

Limitations, next steps and interview pitch

Question 21. What are the most serious limitations?

Two liver-metastasis sections, weak boundary and HGP labels, spot-level cell mixing, incomplete liver-inclusive single-cell reference, co-expression rather than functional signalling, and no patient outcome endpoint.

Question 22. What would you validate first?

First obtain consensus pathologist boundary/HGP scores. Then perform multiplex IF or IHC across signed bands for malignant, pEMT, macrophage, fibroblast, endothelial, hypoxia/glycolysis and damaged-hepatocyte markers.

Question 23. Which perturbations are most informative?

Test macrophage-contact and stromal axes such as CCL18-PITPNM3, SPP1-CD44/ITGAV, TGF-beta-TGFBR2 and CXCL12-CXCR4 in CRC liver organoids, co-cultures or precision-cut liver models. Perturbations should be spatially read out, not only measured in bulk.

Question 24. How would you upgrade the model?

Add pathologist labels, a liver-inclusive scRNA reference, cell2location or RCTD benchmarking, more independent CRLM cohorts and patient-level endpoints. Only then compare the linear baseline with a relational GNN and H&E; foundation embeddings.

Question 25. How is the project reproducible?

The repository contains a single pipeline command, modules for scoring/interface/graph/model/reporting, tests, manifests, small derived outputs, figures and a Streamlit workbench. Large public files remain local under data/raw and are excluded from Git.

Question 26. What did you personally demonstrate through this project?

The project shows end-to-end reasoning: translating pathology into computable labels, designing biological graph relations, controlling contamination and leakage, validating at the correct section level, interpreting a surprising negative result and communicating uncertainty.

Question 27. Give the 30-second pitch.

I built a compartment-aware spatial graph model of the CRC liver-metastasis interface. It separates malignant partial EMT from stromal signal, assigns a signed 300 micrometre boundary, and encodes host-cell, ECM, metabolic and ligand-receptor context as typed relations. The held-out-section baseline reached AUC 0.908, but I treat it as a weak-label hypothesis generator because only two liver sections and no pathologist HGP labels are available.

Rapid-fire numbers

Item	Value
Visium sections	6: four primary CRC and two liver metastases
Spots	22,260 total; 3,439 malignant model spots
Tumour-front spots	1,290
Features	48 local and relation-aware features
ROC AUC	0.908
Average precision	0.857
Balanced accuracy	0.831
Permutation test	50 shuffles; P=0.0196; null AUC 0.495
Matched scRNA reference	56,072 recovered cells across four groups

Safe closing line

> The strongest outcome is not a treatment claim; it is a clear, testable map of which tumour-host regions, cell states and interaction axes deserve pathology review and perturbation next.

Core sources

GSE225857: https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE225857
Science Advances: https://doi.org/10.1126/sciadv.adf5464
HGP consensus: https://pmc.ncbi.nlm.nih.gov/articles/PMC5680474/
Diagnostic reproducibility: https://pmc.ncbi.nlm.nih.gov/articles/PMC6611753/
CRLM spatial fronts: https://www.nature.com/articles/s41416-026-03567-y
Zenodo GeoMx: https://doi.org/10.5281/zenodo.7520789